REVIEW

Very Large Scale Monoclonal Antibody Purification: The Case for Conventional Unit Operations

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Technology development initiatives targeted for monoclonal antibody purification may be motivated by manufacturing limitations and are often aimed at solving current and future process bottlenecks. A subject under debate in many biotechnology companies is whether conventional unit operations such as chromatography will eventually become limiting for the production of recombinant protein therapeutics. An evaluation of the potential limitations of process chromatography and filtration using today's commercially available resins and membranes was conducted for a conceptual process scaled to produce 10 tons of monoclonal antibody per year from a single manufacturing plant, a scale representing one of the world's largest single-plant capacities for cGMP protein production. The process employs a simple, efficient purification train using only two chromatographic and two ultrafiltration steps, modeled after a platform antibody purification train that has generated 10 kg batches in clinical production. Based on analyses of cost of goods and the production capacity of this very large scale purification process, it is unlikely that non-conventional downstream unit operations would be needed to replace conventional chromatographic and filtration separation steps, at least for recombinant antibodies.

Contents

1. Introduction

The production efficiency for recombinant protein manufacture has progressed remarkably in the past 20 years. Advances in cell culture technology such as improved production media and feeding strategies have resulted in increases in peak cell densities (as high as 30 million cells per milliliter) along with extended culture durations of 14 or even 21 days. Maintenance of specific cellular productivities of over 20 pg/cell/day over this period are now common. The combination of these factors can give rise to product titers that are an order of magnitude higher than the typical titers just 10 years ago. Monoclonal antibodies (mAbs), which generally express well in mammalian cell hosts, are often accumulating to 3−5 g/L (*1*); the highest mAb titer this author is aware of was 9.8 g/L achieved with a recombinant CHO line in a 21-day culture (*2*).

In addition, the maximum scale of mammalian cell bioreactors has doubled in the same period. In the 1990s, a bioreactor volume of 10,000 L was considered very large scale; now several companies have installed 15,000 and even 25,000 L bioreactors. The combination of high titer processes produced in these large bioreactors will result in a 50 or even 100 kg batch size for mAbs. Consider this in the context of an early biopharmaceutical product, recombinant Factor VIII, a protein that is difficult to express and stabilize. Currently, three companies are licensed for sale of Factor VIII in the U.S. and Europe; the estimated quantity of protein needed to satisfy the total market of over $1 billion is on the order of 200−300 g of protein. The production of mAbs at the 100 kg scale will certainly tax the limits of purification processes and manufacturing suites. Or will it?

In the near future, purification processes will be designed to handle these very large batches. Conventional unit operations such as centrifugation, low-pressure chromatography, and ultrafiltration are used for all currently licensed monoclonal antibody processes, where batch sizes are more likely to be in the 5−10 kg range. There has been much discussion in the

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Table 1. Very Large Scale cGMP Purification Processes

product	est annual prod (tons)	major producers	source	purification process	sale price ($/g)	licensure date
HSA	500	Talecris CSL Behring Baxter	plasma	Cohn fractionation ultrafiltration	3	1940s
IgIV	80[a]	Talecris CSL Behring Baxter	plasma	Cohn fractionation ultrafiltration	60−70	1980s
insulin	10	Eli Lilly Novo Nordisk Sanofi Aventis	microbial	centrifugation + disruption[d] 6 columns (SEC + RP) crystallization	500	1982
Rituxan	1	Genentech Lonza[b]	CHO	centrifugation 3 columns 2 ultrafiltration steps[e]	4,000	1987
Enbrel	1	Amgen Wyeth BI[c]	CHO	microfiltration 3 columns 4 ultrafiltration steps[f]	4,000	1998
10-ton Mab	10	hypothetical	CHO	centrifugation 2 columns 2 ultrafiltration steps	hypothetical	

[a] Based on ref 30. [b] Produced under contract. [c] Boehring Ingelheim. [d] Based on the Lilly process described in ref 8. [e] Based on ref 31. [f] Based on ref 32.

literature and at technical conferences about the need to develop novel unit operations for very large scale (VLS) processing. Examples of such technologies could include liquid−liquid extractions using two-phase aqueous polymer systems, expanded bed adsorption, crystallization, simulated moving bed chromatography, continuous processing, and precipitation (*3*). In some cases, this work is motivated by the perception that conventional chromatography and other unit operations will become limiting for VLS processes.

Each of these novel unit operations offers certain advantages over conventional unit operations, but not all will be amenable to a VLS process. In fact, conventional chromatography and ultrafiltration will likely be sufficient to process these VLS batches. To argue this point, this paper will elaborate details of a conceptual design for a VLS cGMP (current Good Manufacturing Practice) mAb purification process. An analysis of the direct cost of goods (COGs) for such a process suggests that reduction in raw material costs will not be a significant driver for considering alternative technologies. Additional process efficiencies, improvements in scheduling, and tandem operation of steps could enable even greater production scales.

2. VLS cGMP Protein Purification Processes

It is informative to review current cGMP processes, which produce tons of protein products per year. From these examples will come insights into process and facility design that will be relevant to the 10-ton mAb conceptual design (see Table 1).

Two products derived from human plasma are produced at scales greater than 1 ton per year. Human serum albumin (HSA) is used for treatment of shock and blood loss, and the total market is estimated to be approximately 500 tons per year. Immune globulin (intravenous), called IgIV, is a polyclonal mixture of IgG antibodies and is licensed for several indications, including treatment of primary immune deficiency. The IgIV market is estimated to be approximately 80 tons per year. For both of these products, production capacity is split between many plants in operation throughout the world, owned by either large commercial multinational fractionation companies or smaller, publicly funded blood services processing domestically sourced plasma in national plants. Although precise production capacities are not available, the world's largest plasma fraction plant (Talecris in Clayton, NC) is estimated to process 3 million L of plasma per year. Using typical yields for HSA (*4*), this single plant would generate approximately 100 tons of HSA per year. At this production scale, HSA is truly a commodity product, which is reflected in the sales price of approximately $3 per gram of HSA. IgIV commands a much higher price (about $60−70 per gram) for certain indications. The total IgIV production capacity is also distributed between many plasma fractionation plants. It is possible that the largest production plants may be producing around 10 tons of IgIV per year, which is coincidently the production capacity of the conceptual design for the VLS recombinant mAb process to follow.

For recombinant products, only a few are produced at ton scale: insulin, Rituxan, and Enbrel. Insulin production is split between three principal suppliers: Lilly, Novo Nordisk, and Sanofi-Aventis. If the production capacity for both recombinant human insulin and the recently approved insulin homologs is combined, the total market is approximately 10 tons. These three companies use microbial hosts for insulin synthesis (Lilly and Sanofi-Aventis use an *E. coli* host, and Novo Nordisk uses *S. cerevisae*). Rituxan, a monoclonal IgG used for treatment of non-Hodgkin's lymphoma is the largest volume mAb on the market today, with a production scale of about 1 ton per year (Avastin is likely to become the largest volume recombinant mAb soon, and Herceptin production is also close to this scale; both were developed by Genentech, and use similar purification processes). Rituxan is produced at Genentech's Vacaville site, and a second plant at Lonza Biologics has been brought online. Enbrel is a translational fusion of the constant domains of an IgG heavy chain and an extracellular receptor domain for tumor necrosis factor (TNF)-α and is licensed for treatment of rheumatoid arthritis as well as several other indications. It is currently produced at three sites: Amgen's Rhode Island plant, Wyeth's Grange Castle facility in Ireland, and under contract at Boehringer Ingelheim in Biberach, Germany. Both Rituxan and Enbrel use recombinant CHO cells as a production host, require approximately 1 ton of product per year, and thus represent the largest production scale for cGMP proteins derived from mammalian cell hosts.

It is clear that only a few cGMP protein production plants in operation today even approach a capacity of 10 tons per year. Those that do (HSA and IgIV) typically employ the Cohn fractionation process and do not rely on chromatography for purification of these products. (There are some plasma-derived immunoglobulin products that have adopted chromatography, including a very large processing facility in Australia (*5*); however, the majority of IgIV processes are based on the Cohn process (*6*, *7*)). The Cohn process employs selective precipitation steps that manipulate pH, temperature, ethanol concentration, and ionic strength. Additional unit operations for plasma-derived products include microfiltration, ultrafiltration, and centrifugation. The recombinant proteins are all produced by purification processes that rely heavily on chromatography, although insulin production also uses crystallization. How is it that chromatography has not been used yet for the very largest cGMP protein processes? Does chromatography have some limitation that would make it impossible or unfeasible to process at a 10-ton scale? From this pattern of applied technologies, it could appear that chromatography will not be suitable for VLS production; this may explain why there could be a perception that capacity limitations are a driver for alternate technologies.

The history of the plasma processing industry may explain the preference for these specific unit operations. When the Cohn fractionation process was developed and transferred to multiple production sites, the chromatographic resins available at the time (1950s) did not have the high performance that current resins enjoy. Resins were soft and compressible, which made scaling to large columns difficult. Dynamic binding capacities were low, and therefore large batches could not be processed in a single cycle. Once plants were built to run Cohn-based processes and multiple licenses were granted for production of the many products that can be derived from a fractionation scheme, there would be a reluctance on the fractionators' part to make wholesale changes to the production process. Changing a purification process step in the trunk of the Cohn process could have unintended impact on other products derived from subsequent process steps. In some cases, chromatography has moved into the fractionation processes, but it is typically used for products with relatively lower production scales (Factor VIII, etc.). It is interesting to speculate what changes to the plasma fractionation processes would emerge if the plasma fractionation industry did not have 50 years of experience with a Cohn fractionation process platform and were starting over with today's technology. This author believes it would be likely that there would be more chromatography steps, driven by increases in yields, elimination of the use of ethanol, and avoidance of sub-zero temperatures in processing. There are processes under development today that would employ multiple chromatographic steps, reflecting the advances made in chromatographic media in recent decades as well as taking advantage of the development of novel affinity ligands for specified targets (*8*).

It is more likely that chromatography will cover the VLS processes being developed for recombinant mAbs, and less likely that precipitation-type unit operations will be used. Support for this argument may be gathered from a review of the insulin manufacturing processes. The Lilly process begins with an *E. coli* expression system that generates proinsulin inclusion bodies (*9*). Following disruption, clarification, dissolution, and refolding, the product is purified by six chromatography steps including reversed phase (RP), ion exchange, and size exclusion chromatography (SEC). Some of these modes are inherently low capacity separation methods, yet the production plant for insulin is producing several tons per year and selling the product for approximately $500 per gram. In comparison to this process, a CHO-based mAb purification process is much simpler. The product is secreted in a correctly folded form and may be captured with an affinity chromatog-

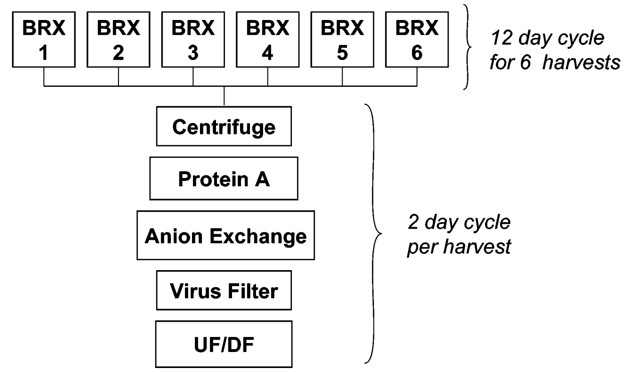

Figure 1. Process flow diagram for the 10-ton mAb purification process.

raphy step run at high capacity that can provide a product of >95% purity. The polishing steps are typically ion exchange steps with even higher capacities and can enjoy large selectivity factors based on the fact that the humanized mAbs are generally more basic than the acidic host cell proteins.

3. Facility Design Considerations

The facility and purification process described in this paper would produce 10 tons of a mAb (60 kg per batch) derived from a CHO cell culture process. The very high production capacity will be enabled by a greenfield production plant with adequate sizing of the facility's process suites, equipment, raw material warehouse, and bulk drug storage. As a base case, the factory would be designed expressly for a single product. The cell culture process suite would be equipped with six identical production bioreactors, each of 15,000 L working volume. The production bioreactor growth and production phase would be completed in approximately 10 days, to allow the bioreactor to complete inoculation, growth/production, harvest, and CIP/SIP phases on a 12 day cycle (this incorporates a 2 day bioreactor turnaround period). The downstream process suite is designed to purify a single harvest at a time and scheduled to process a new batch every other day (Figure 1). There are several examples of currently operating biopharmaceutical plants that share this design principle of a single purification train servicing multiple large-scale bioreactors.

The mAb titer in each harvest would average 5 g/L, thus providing 75 kg of product in each harvest. Assuming an 80% purification yield, each batch would thus generate 60 kg of purified bulk drug substance (BDS). In a full production year, a 10-ton capacity would require 167 harvests in 334 days of production. Assuming a month to build up an inoculum, the full year would be booked with production and a few weeks of shutdown (assuming a rolling shutdown between inoculum, production bioreactors, and downstream suites). A full year of production with no shutdown would yield 182 harvests, or a production capacity close to 11 tons.

This scale of purification process operations exceeds the production capacity for all but the IgIV and HSA processes described above. The plant would therefore provide what would currently be the world's largest recombinant product mass per year, from a single purification train with only two chromatographic columns! The amount of product generated is remarkable; if the BDS was concentrated to 100 g/L, the plant would yield 100,000 L of BDS per year. The capacity limit could be shifted to the drug product manufacturing process, as the volume of BDS is high enough to potentially challenge the logistics of the supply chain and storage facilities. (Consider the challenge of frozen storage of the full volume of BDS at −50 °C, or even

−20 °C compared to the benefits of liquid storage.) Will the production bottleneck be shifted to the fill-finish operations?

If a higher capacity cell culture process were to be developed, the facility design could easily be modified to accommodate additional throughput. If the product titer were to double to 10 g/L, a second purification suite could be added to the plant, and the harvest split between the two suites. While this may be considered an inelegant solution that does not take advantages of the economies of scale from running a single train (which is still possible and is described in Section 7), it would offer great flexibility for the situation where the six bioreactors are used to produce two products (at 5 g/L titer) simultaneously. This would allow tailoring of the plant's production capacity to the shifting demand ratio for the two products, which would be of great benefit for managing two product supply chains. Obviously, the use of a single purification platform technology for both products would be useful in managing such a multi-product facility.

An alternate scenario is one in which an existing plant is retrofitted for the 10-ton process. In this case, there could be several factors that could eventually limit the production capacity if these constraints cannot be relaxed through retrofit or significant capital investment. These potential limitations are addressed below.

• **Space.** The floor space needed for purification may be inadequate or configured improperly. Earlier processes that did not handle large harvest volumes, did not require chromatography or ultrafiltration skids for multi-kilogram batches, or had several small suites each serving a single bioreactor may result in a facility layout that would require complete remodeling.

• **Solution Preparation.** The solutions needed for the purification process may exceed the production capability of the existing solution preparation suite. Large tanks or solution concentrates may be required in order to produce the tens of thousands of liters of solutions needed for the processing of a 60-kg batch.

• **In-process Storage Tanks.** The large mass of product being processed by the purification train would require handling significant volumes of in-process product pools (chromatography step elution pools). For a 60-kg batch, an in-process pool with a low product concentration, say 5 g/L, would require a 12,000 L holding vessel. Most purification suites do not have in-process tanks that are as large as the production bioreactor! Current technology using disposable bags to hold in-process pools would be limited to approximately 2,000 L per bag.

• **Water for Injection.** The consumption of water-for-injection (WFI) required for the purification process in a VLS plant can be enormous. The volumes of solutions are small compared to the overall WFI consumption, as WFI is used for CIP of equipment and transfer lines in addition to the flushing of filters. Existing facilities may require additional WFI stills and improved storage and distribution systems to handle the increased capacity required by the purification process.

• **Storage and Distribution of BDS.** While not a technical challenge, the logistics of handling, storing, and shipping 600 L of BDS every other day are not trivial and may command additional space and specialized transfer equipment.

These are factors which may result in a limitation of production capacity, or bottleneck, for an existing facility. In the conceptual process described in Section 4, these pressures are alleviated somewhat by the efforts of process intensification which result in a highly efficient purification process. The number of unit operations is minimized (specifically, only two chromatographic steps are used). This reduces the process footprint, eliminates several solutions and raw materials needed for additional chromatography steps, does away with the associated in-process pool vessels or bags, reduces labor costs, and provides a more rapid conversion of harvest to BDS. The use of modern, high capacity resins enables in-process pools to be highly concentrated, which minimizes the volume of in-process storage tanks, and also shortens the processing time of the subsequent step.

Table 2. Details of 10-Ton mAb Purification Process Unit Operations

Centrifugation and Harvest	
centrifuge flowrate	3,750 L/h
pad filter volumetric challenge	200 L/m^2
0.2 μm Prefilters	
volumetric challenge (varies with stream)	100−1000 L/m^2
Protein A Chromatography	
column diameter	200 cm
bed height	50 cm
bed volume	1,500 L
[mAb] in load	5 g/L
linear velocity	300 cm/h
dynamic binding capacity	50 g/L
elution volume	2.5 CVs (3,750 L)
Anion-Exchange Chromatography	
column diameter	120 cm
bed height	25 cm
bed volume	320 L
[mAb] in load	20 g/L
linear velocity	150 cm/h
column load capacity[a]	250 g/L
pool volume	4,710 L
Virus-Retaining Filtration	
membrane area	8 m^2
average flux	40 L/m^2 h
[mAb] in load	20 g/L
Ultrafiltration/Diafiltration	
[mAb] in load	18−19 g/L
membrane area	100 m^2
average flux	30 L/m^2 h
[mAb] in load	18 g/L
diafiltration volume	10 diavolumes
target mAb concentration	100 g/L

[a] The AEX step is defined not by the product's dynamic binding capacity, as in the ProA step, but the total product loaded; only a fraction of the product binds under weak partitioning conditions.

4. Description of A Two-Column mAb Purification Platform Process

4.1. Process Flowsheet. The purification process flowsheet is shown in Figure 1, and details of the processing operating conditions are given in Table 2. The purification process is based on cGMP processes that have been run at Wyeth for several mAbs and uses unit operations typically employed for mAb purification currently (*10*, *11*). Wyeth's clinical production facility has both 2,500 and 6,000 L bioreactors; several different products have now generated batches of over 10 kg from single harvests. While not all aspects of this process have been tested at scale (the 50 cm bed height of the capture column, for instance), the scale-up factor for the 60-kg batch is rather modest.

The CHO cell bioreactor would be harvested by a continuous disc stack centrifuge, and the centrate passed through a pad filter before collection in the harvest vessel. Centrifugation has become the preferred unit operation for clarifying CHO cultures at large scale over the past decade (*12*, *13*). The harvest would require 4 h of processing time for an average flowrate of 3,750 L/h. The centrate would pass through a pad filter prior to entering the holding tank. There is nothing unusual about this harvest operation, and the centrifuges and pad filter equipment needed to handle this stream and generate clarified centrate at this flowrate are currently in operation at several biopharmaceutical plants. The prefilters used in the purification train are intended to reduce small levels of particulates, and the volumetric prefilter challenges used for the design basis range from 100 to 1000 L/m^2, depending on the extent of the particulate burden as well as the processing time, which results from the combination of filtrate volume and flowrate. Different areas are used for the various streams, but the prefilter material itself is the same for all applications.

The clarified centrate is then loaded directly onto a Protein A (ProA) chromatography column, where impurities such as host cell proteins, nucleic acids, and cell culture media components are removed. Direct capture of mAbs from filtered centrate is a common practice in the biopharmaceutical industry and presents no significant technical issues. The ProA column is designed to handle a 75-kg challenge in a single cycle of operation, which would be unique. Assuming a dynamic binding capacity of 50 g/L, the column volume required is 1,500 L, which would be accommodated by a 2 m diameter column packed to a 50 cm bed height.

This ProA column dynamic binding capacity is quite high, close to the upper limit reported value for currently available resins. There are several factors that make this 50 g/L target a reasonable goal. First, equilibrium binding capacities of 60–65 g/L have been reported for some ProA resins (*13*, *14*). Second, the load concentration will be much higher than most current feedstreams, and this should result in a modest increase in dynamic capacity. Third, the column design calls for a 50 cm bed height, in combination with a relatively long (10 min) residence time. Both of these factors will increase the dynamic binding capacity over more typical operation (25 cm bed heights and 4–8 min residence times), and it is likely that the dynamic binding capacity would be within 85–90% of the equilibrium binding capacity. (Using shorter bed heights of 30 cm and faster residence times of 6 min, dynamic binding capacities of greater than 40 g/L have been achieved at Wyeth.)

A bed height of 50 cm is an unusual feature of this ProA column, although bed heights for process ion-exchange columns exceed this height for some processes. It is quite likely that the two Protein A media in common use today could accommodate this bed height, as their matrix backbones are either highly cross-linked agarose or controlled pore glass (*15*). The glass medium should not offer any problems with compression or back-pressure, although currently the 50 g/L dynamic binding capacity for this matrix is not achievable. The highly cross-linked agarose medium provided data on resin hydraulics indicating that a 2 m diameter column packed to a bed height of 50 cm would have a critical velocity of approximately 450 cm/h, which is slightly faster than the flowrate proposed here. Should bed compression become an issue, reducing the flowrate would relieve the pressure limitation, at the expense of a modest increase in processing time.

The ProA elution pool pH would normally be below 3.8, or adjusted with acid if required, which provides assurance of viral clearance by inactivation of many enveloped viruses, including the retrovirus models of the CHO cell Type A non-viable retrovirus-like particles (*16*). The product could be eluted in approximately 2.5 column volumes, generating a 20 g/L product pool. This high concentration helps reduce the process footprint and also the requirements of the purification process for storage of in-process pools. For the two-column purification train described here, the ProA pool volume also impacts the sizing of the downstream anion-exchange, virus-retaining filtration, and ultrafiltration steps.

The second chromatography operation is a polishing step using an anion exchange (AEX) resin to further reduce impurities to the purity target in BDS. The AEX resin exploits the fact that the mAb is more basic than these impurities, which bind more tightly to the resin. The ProA peak is adjusted to the appropriate pH and then loaded onto the pre-equilibrated column. The column is operated in an isocratic mode that is called weak partitioning chromatography (WPC) (*17*). In WPC, the load solution conditions are chosen to allow a significant amount of product to bind to the resin (1–20 g/L), much higher than the typical flow-through operation of an AEX column used for mAb processing (*18*). For these more stringent binding conditions, impurities are removed to a greater extent than flow-through conditions. Under optimized conditions, over 4 logs of CHO proteins may be removed, along with substantial clearance of leached Protein A (>2 logs), nucleic acids (>3 logs), virus (>5 logs for retroviruses), and aggregate (up to 20-fold reduction in some cases). An additional benefit of the WPC operation is that very high load challenges are possible; the design basis for this step would load 250 g of product per liter of resin volume. Despite the elevated product binding to the resin, step yields are acceptable because the load volumes are high, and a short isocratic wash (approximately three column volumes) is used to recover most of the bound product. This allows the AEX column to be much smaller than the ProA column and to employ a more modest bed height for lower back-pressures and faster operation. The single polishing step needed for this process offers many advantages, some of which will be summarized below in comparison to the standard three-column processes in common use today.

The AEX product pool is then passed through a normal flow virus-retaining filtration (VRF) ultrafilter with a pore size small enough to ensure clearance of parvoviruses (*19*). The filter area is chosen on the basis of a moderate volumetric challenge (400 L/m^2) (*20*) and would be operated with a module that does not result in a significant flux decay (*21*) that could give rise to variable processing time. Other options for filter sizing for this step and their impacts are described in the section on COGs below. The protein concentration in the load to the VRF step would be higher than typical operations currently, but these conditions have been shown to yield acceptable back-pressures and stable operation in laboratory studies and should not pose a risk to process robustness. Hongo-Hirasaki et al. have processed a 30 g/L mAb solution with a parvovirus filter (*22*), well above the 20 g/L concentration used here. (The process flowrate was relatively low for this 30 g/L feed, but the next highest protein concentration of 10 g/L maintained a flowrate of almost 40 LMH.) The VRF step flowrate could be decreased from the target 40 LMH of the design basis if needed without significant impact to the overall facility productivity.

The final step in the purification process is an ultrafiltration/diafiltration operation that concentrates and formulates the product, generating the BDS pool. An average flux of 30 LMH is assumed, and the process employs a 10-fold diafiltration volume at 50 g/L before the final concentration to the 100 g/L target. The protein concentration in the feedstream to this step would be around 16–18 g/L, as the previous two steps do not cause much dilution of the ProA pool. The corresponding UF/DF load would be the largest pool of the purification train (excepting the harvest) and would be approximately 4,000 L.

One version of an architectural layout of the purification process is shown in Figure 2A. The harvest vessel, centrifuge,

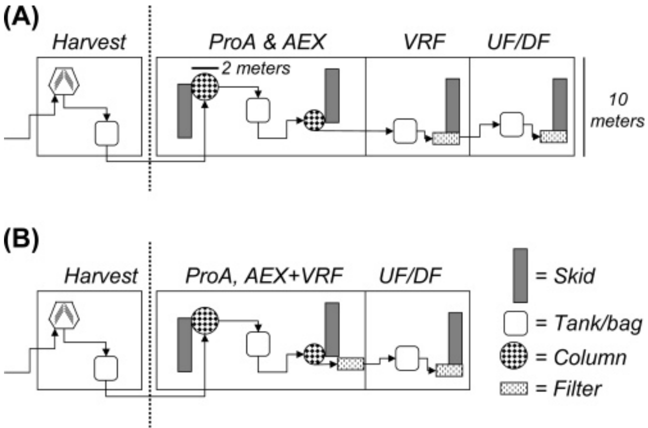

Figure 2. Architectural layouts for the purification process: (A) base case, (B) combination of the AEX and VRF steps.

and pad filtration equipment are located in the cell culture suite. The clarified centrate would be transferred to the purification suite through transfer piping. There, the two chromatography steps could share a single room, where solutions would arrive from the solution preparation suite through transfer panels, so no large storage vessels would be needed. The ProA and AEX step product pools could be held in modestly sized tanks or shared between a pair of 2,000 L disposable bags. The VRF and UF/DF steps would have dedicated rooms to allow for clear segregation of pre- and post-virus clearance steps, as well as to provide for the heightened environmental controls required for the area where BDS is generated. The footprint for such a purification process would easily fit in a 4,000 ft^2 area (the narrow dimension of the suite layout shown is just 10 m), which is only a modestly sized process suite. While pad filters for the process skids are not shown in this layout, only the neutralized ProA pool might require a pad filter on a separate skid for clarification; all other 0.22 μm filters would be part of the chromatography or ultrafiltration skids.

The process yields are expected to be high for the purification train. Centrifugation losses can be minimized by buffer flushes before discharge and should exceed 95%. ProA chromatography step yields are typically 95%, as well. The AEX step has a very high load challenge and includes a 3 column volume wash to recover the product that is bound to the resin. In our experience, a well-designed WPC step exceeds 95% recovery. Finally, the VRF step should give nearly quantitative recovery, and the final UF/DF plus sterile filtration step yields would approach 98%. The overall process yield assumption of 80%, therefore, is a very reasonable value. The clinical processes at Wyeth that have been run using a similar two-column process and which generated 10-kg batches have overall process yields averaging roughly 80%.

4.2. Two-Meter Protein A Column. The proposal to operate a 2 m ProA column with 1,500 L of resin bears some discussion. While there are clear advantages to a single cycle of operation, the more common strategy for processing large volumes of clarified centrate across a ProA capture column is to cycle a smaller column several times. The decision to cycle once or multiple times is often one that is determined by consideration of operational factors and is not based on technical limitations that would preclude one or the other scenario. (An exception to this would be the case where the product in the centrate was unstable, and rapid processing would be necessary to preserve yield or prevent the generation of undesirable product-related impurities.) There are many benefits of column cycling, which will not be reviewed here. Reducing the column diameter and increasing the cycle number would not be difficult (a 1.4-m column would be cycled twice, and a 1.0-m column would require four cycles). These processing options are very reasonable and would not give rise to an overall purification process duration exceeding the 2 day cycle of the bioreactor harvests. The decision of whether to run single or multiple cycles is not a critical element of the design basis, nor will it limit the capacity of the 10-ton process, but instead can be taken on a case-by-case basis.

The high dynamic binding capacity of the ProA step, as described above, is the result of the tall bed height, high load concentration, and long residence time. The benefits of this design should be maintained no matter which column cycling scheme is chosen, as the high concentration of product in the ProA pool has additional benefits as previously described.

The size of the 2 m diameter column is not unprecedented, as this scale of chromatography has been employed for multiple cGMP protein purification processes in several manufacturing facilities and has been characterized in column packing studies (*23*, *24*). Some manufacturers of chromatography columns offer 2-m columns in their catalog, as an “off the shelf” item. The capital cost of the equipment needed for this step is not excessive (the chromatographic column would be around $350,000, the control skid will be about the same cost, and a packing skid would be less than $100,000). This $800,000 total is small in comparison to the bioreactor capital investment, as each individual bioreactor and control skid may cost approximately $2 million.

One concern of single cycle operation is that the resin cost of the 1,500-L column is quite high. At current prices for ProA resins, this would amount to around $12 million of resin (in addition, most facilities operate with enough resin in inventory to allow a back-up column to be pressed into service, thus doubling the total resin inventory!). The microbiological control of the column will be critical, to avoid loss of the resin due to contamination. Resin cleaning could also be problematic, as the regeneration solution would need to ensure adequate removal of any bound impurities following product elution and would require even flow distribution across the large 2-m column as well. Despite these concerns, the technical challenges for use of ProA resins for direct capture of mAbs from mammalian cell culture media at room temperature appear to largely have been solved.

4.3. Weak Partitioning Chromatography. While a full explanation of WPC conditions is beyond the scope of this paper, some comments are worth noting. The use of WPC with AEX as the only polishing step contributes to the efficiency of the purification train, preserving the isocratic operation of typical flow-through AEX steps, as well as a minimal dilution of the product pool. The high column loading capacity is a benefit in comparison to standard flow-through operation. Despite the significant levels of product binding to the resin during operation, process yields for WPC steps exceed 95% as a result of the high loadings and short washes. Using well-established Design of Experiments (DOE) process characterization methods (*25*), WPC steps used in several processes have been demonstrated to be robust.

The use of a single polishing step is not a critical element of the argument that conventional chromatography steps will not be limiting for the 10-ton process. Adding a third chromatographic column, especially for a greenfield manufacturing facility, will not result in capacity limitations. The capacity analysis for retrofitting of an extent facility, however, could be impacted by the need for a third chromatographic step.

Table 3. Purification Process Costs and Sensitivity Analysis, COGs, and Solution Consumption

base case purification process costs per unit operation[a]	$/g	top 5 raw materials (base case)	$M/yr
harvest (centrifugation and clarification)	1.02	ProA resin	13.1
ProA	2.03	0.2 μm prefilters (all)	5.6
AEX	0.35	VRF membrane	4.4
VRF	0.50	buffer used in harvest and ProA steps	3.6
UF/DF	0.12	ProA step wash salt	3.5
total (direct mat'l COGs − purification)	4.02	purification process costs (total)	40.2
total COGs estimate[b] (drug product)	$/g	sensitivity analysis: % increase over base case purification process costs	%
upstream (medium cost of $8/L)	2	WFI cost doubles to $200/kL	+4
purification raw material costs	4	ProA lifespan reduced to 50 cycles	+98
facility depreciation	5	ProA capacity reduced to 25 g/L	+61
staff salaries	5	AEX capacity reduced to 125 g/L	+3
fill-finish cost per vial	10	VRF area increases 4X to 32 m^2	+32
total COGs (1 g/vial)	26	add a third chromatographic step[c]	+29
total COGs (0.1 g/vial)	≈12	combine six sensitivity scenarios above	+465
		drug substance purification cost	$22.60/g
solution consumption	(L/g)		
base case	1.5		
add a third chromatography step[c]	2.5		
combine six sensitivity scenarios	4.7		

[a] All raw material costs for the unit operation are included (resins, membranes, 0.2 μm prefilters, chemicals). [b] The other components of the full cost of goods estimate shown here and the basis for their values are described in the text. [c] The third chromatographic step would run in a bind and elute mode, have a 50 g/L dynamic binding capacity, and provide a 90% product yield (all other design features are the same as the AEX step in the base case process).

5. Cost of Goods Analysis

5.1. Description of Model. The production costs for the 10-ton mAb purification process were estimated using both an Excel spreadsheet and commercially available process modeling software, which gave similar results and were used to cross-check calculations. The direct cost estimate was restricted to direct costs only and did not account for capital or facility depreciation, labor, QA/QC, etc. These direct costs can be considered to be accurate estimates and are based on assumptions of purification process parameters that are listed in Table 2. Raw material costs (chemicals, resins, membranes) were based on 2006 pricing information obtained from the internal raw material sourcing group at Wyeth.

Important assumptions for the model include the maximum resin and ultrafiltration membrane lifespans. A 200-cycle lifespan was chosen for the ProA resin, and a 100-cycle lifespan for the AEX resin and membranes; 200 cycles of use for ProA would be considered very good by current standards, although longer lifespans have been reported (*26*). The resultant cost of the ProA resin is close to $1 per gram of mAb, dispelling the perception that ProA chromatography will be a major expense for mAbs. It should be noted, however, that ProA costs for clinical production campaigns are much higher (where 1−20 cycles of use would be more common), as is the case for commercial mAb production where steady year-round production is not necessary (consider the campaigning scenario described in Section 8). These economic impacts are important drivers in the decision regarding the number of ProA cycles used per batch.

The cost basis for the WFI used for process solutions is assumed to be $0.10 per liter for this analysis. Cost estimates for WFI vary widely, reflecting the wide range of true costs, which depend on scale of operation, as well as the uncertainty of all factors to include in the cost estimate. It is quite likely that the operation of the 10-ton mAb facility will provide economies of scale that will result in actual WFI costs being much lower; the economic analysis presented here, therefore, could be considered as a conservative estimate. The two-column process uses very low volumes of process solutions (1.5 L/g of purified mAb), and the minimal number of unit operations also require less WFI during CIP/SIP, which diminishes the sensitivity to WFI costs.

5.2. Model Output. The direct raw material cost of the purification process alone is approximately $4 per gram of mAb drug substance produced (Table 3). The top five raw material costs are the ProA resin, the 0.2 μm prefilters (seven are used in the process flowsheet), the VRF membrane, and two chemicals used in the harvest and ProA step. The cost estimate for the cell culture process is not included here, but a rough estimate for a chemically defined medium costing $8 per liter would give rise to approximately $2 per gram of mAb (based on a 5 g/L titer and 80% purification yield).

The direct raw material costs for the purification process are broken into the various unit operations in Figure 3. The cost for each unit operation combines the resin or membrane cost, all raw materials used in that step, and the 0.2 μm prefilters. The ProA step is the most expensive and is driven primarily by the price of the chromatographic resin. The harvest operation is second and has large costs associated with the pad filter used to clarify the centrate, as well as the expense of the buffer used to adjust and maintain the centrate pH. The VRF step is also expensive, solely due to the cost of the ultrafilter and the fact that it is a single-use device.

This value for the direct raw material costs is comparable to other estimates that have been reported in recent technical conferences from biopharmaceutical companies or vendors supplying critical raw materials. The total COGs will be much higher once facility depreciation and labor costs are factored in. One estimate for these additional costs is provided here and summarized in Table 3. If the 10-ton mAb facility cost $500 million to construct (which would be approximately $5,000 per liter of bioreactor capacity, which is at the upper range of a 2001 review published by Farid (*27*)), this would result in a $50 million annual operating cost assuming 10-year straight line depreciation or approximately $5 per gram of mAb. A staff of 250 employees needed to operate the plant would also draw approximately $50 million per year in salaries and benefits. Fill-finish costs may account for approximately $10/vial. A smaller 0.1 g dose vial would therefore have a total production COGs of approximately $22, whereas a 1 g dose would cost $26. It is worth noting that the cost of the fill-finish operation would

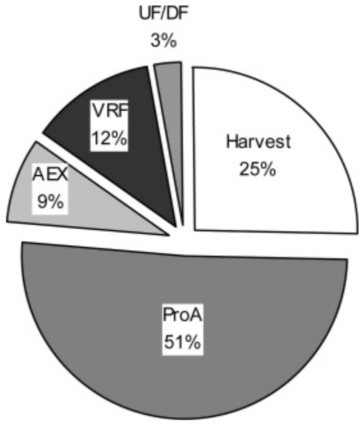

Figure 3. Cost of goods breakdown by unit operation.

exceed the cost of the drug substance for the smaller dose, a situation more commonly encountered with small molecule pharmaceuticals.

Note that if the cell culture titer were only 0.5 g/L, as was typical 10 years ago, the annual production capacity of the plant would be just 1 ton. The direct raw material costs for the same two-column purification process would still only be approximately $4 per gram (assuming the ProA capture capacity is still 50 g/L), but the facility depreciation and labor would now be 10-fold higher on a per-gram basis (approximately $100 per gram). In addition, the cost of the cell culture medium would also increase by a factor of 10 (in this estimate, it would rise from $2 to $20 per gram). Combined with the consistent fill-finish cost of $10/vial, a 1 g dose would cost approximately $134 per gram (see Table 4).

This model only includes detailed calculations for the direct raw material costs for purification steps. The reduction in the number of process steps afforded by the two-column process will also reduce the number of process operators, validation engineers, QA/QC staff, etc. needed to operate the plant. Less equipment would translate into lower capital investment and lower overhead. The benefits of the process intensification will also be seen in the reduction of these and other indirect costs. While the values that were assumed for the fill-finish costs, facility capital investment, and staffing levels were round numbers, they are believed to be conservative values that would not lead to significant underestimates of the product COGs.

5.3. Sensitivity Analysis. A sensitivity analysis identifies several areas of focus for purification raw material cost reduction. (Even though the direct raw material cost of the purification process is only $4 per gram for the 10-ton process, the plant still incurs $40 million in raw material costs per year.) The increases in purification direct raw material costs for six different scenarios are summarized in Table 3. The WFI cost is not very large (doubling the WFI cost only adds $0.16 per gram), but this estimate does not account for the larger use of WFI in CIP/SIP. However, the conservative cost estimate of $0.10 per liter for WFI costs for the base case is likely to give an elevated estimate of raw material purification cost, which offsets the inability to estimate the full volume of WFI needed to run the plant. The operation of the ProA column markedly influences the process economics; a reduction of the cycle number from 200 to 50 would double the purification cost. A lower column load capacity will have the same effect, in proportion to the additional volume of resin required per year. The AEX step is not a major cost driver, and reducing the load challenge 50% to a more modest 125 g/L value has little impact on the total purification raw material costs. The VRF step is the longest step in the process (approximately 10 h). To decrease the processing time and potentially increase plant output, the membrane area could be increased. A 4-fold increase in the membrane area would make the VRF membrane the single largest raw material cost. Adding a third chromatographic polishing step operated in the bind and elute mode would increase the total purification raw material costs approximately 30%. Although this value would not seem to be very significant (an additional $1.20 per gram would certainly be small compared to the total drug substance and drug product COGs), there are other impacts in facility layout and solution consumption that should be considered. (The solution volumes required for a process using a third chromatographic step would increase 67% to 2.5 L/g.)

Table 4. Breakdown of Costs for Various Process Scenarios

	cost in $ per 1 g vial of drug product				
	cell culture	purification	fill-finish	depreciation and labor	total
base case (10 tons)	2	4	10	10	26
low titer (0.5 g/L) (1 ton)	20	4	10	100	134
3-column purif (10 tons)	2	23	10	10	45
combined low titer and 3 column (1 ton)	20	23	10	100	153

The final entry in the sensitivity analysis is a combination of all six sensitivity scenarios, which would be representative of many mAb purification processes developed in the past decade. The factors are all independent, and therefore the purification raw material cost estimate increases over 5-fold to approximately $23 per gram. The solution consumption increases over 3-fold, from 1.5 to 4.7 L/g. The ProA step costs now dominate, accounting for 77% of the total purification raw material costs. The switch from a current mAb purification process that employs all of these scenarios to the intensified two-column process described here would decrease the direct purification raw material costs by over 80% and reduce solution consumption by nearly 70%.

The sensitivity analyses highlight the importance of several areas of process development or optimization for the purification process. The ProA step should be pressed to maximum loading capacity, which can be provided by long residence times, taller columns, and hopefully, the development of new resins with greater capacities. The ProA column lifetime is also critical and should drive efforts to extend column lifetime through conditioning the column feed to reduce proteolytic degradation, minimizing ligand loss during column regeneration, or the development of more stable Protein A ligands or mimetics. A reduction in the VRF filter area would decrease purification costs, but the long processing time could become a bottleneck. Maximizing the membrane flux at high protein concentrations would be valuable, perhaps through operation at elevated temperatures. A multi-use VRF module would also reduce the total purification raw material cost, and development of validated cleaning protocols to allow reuse for even just four or five cycles would provide most of the benefit of VRF module reuse. The reduction in process solution volumes (and thus WFI consumption) are not a major cost savings but instead factors more strongly into the ability of the process to fit into existing facilities. Perceptions of WFI costs being an important economic driver were likely based on elevated cost estimates for WFI combined with processes that had many-fold higher solution consumption rates. The high cost of the 0.2 μm prefilters can

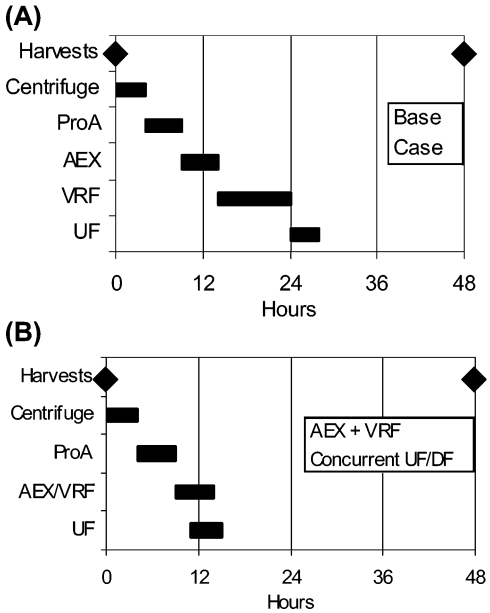

Figure 4. Production schedule for the purification operations: (A) base case, (B) combination of the AEX and VRF steps along with concurrent UF/DF.

be reduced if filters with higher capacities and flowrates could be identified.

One estimate of the COGs for some current mAb production processes is generated by combining the cell culture and purification changes (Table 4). Note how the ratio of purification, cell culture, and fill-finish costs shifts among the four scenarios shown. The combination of both titer reduction (from 5 to 0.5 g/L) and alternate purification scenarios describes a process that is similar to several used for the production of large volume cGMP mAbs today. Here, the drug substance COGs per gram would be $153. Chadd and Chamow estimate the COGs to be $300−1,000 for a mAb in their 2001 publication (*28*). The transition from the case described by Chadd and Chamow (a plant with a 100-kg production capacity from a 0.5 g/L titer cell culture process, using an unspecified purification process) to the 10-ton mAb facility and process described here would be predicted to provide almost a 95% reduction in COGs. This dramatic improvement is the direct result of the great advances provided by process development groups focusing their efforts on continued process improvements and optimization.

The motivation for use of alternative technologies based on COGs reduction should consider the actual direct raw material costs of the two-column purification process. The $4 per gram estimate of the purification process raw material cost is very low, and if a miraculous alternative technology was able to bring this cost down to a smaller value, it would still not have a major impact on the overall COGs, which includes depreciation, labor, and fill-finish operations. This analysis suggests that COGs will not be a major driver for unconventional purification unit operations for VLS processes.

6. Capacity Analysis

6.1. **Scheduling.** An idealized production schedule for the 10-ton mAb purification process is shown in Figure 4A, where the duration of each unit operation reflects the amount of time the product spends in that process skid, based on the process parameters shown in Table 1. The total process time of 27 h fits comfortably within the 48 h cycle between successive harvests from the bioreactor suite. (An actual production schedule would include other activities besides those that deal solely with product manipulations, such as setup time and CIP/SIP cycles, and would provide detail on the overlapping activities of contiguous unit operations.) This schedule allows for only one batch of product to be processed in the purification suite at any given time and minimizes the potential for operational errors associated with multiple unit operations occurring simultaneously on more than one batch. In an alternative production schedule using only two shifts of operation (not three), the chromatography steps could be completed on the first day, and the VRF and UF/DF steps could be completed during the second day.

The ProA column can easily be cycled up to four times and still fit within the 48 h harvest cycle, although this process would have little slack time to recover in case of process or schedule upsets. The slowest unit operation in the base case design is the VRF step. The VRF process time can be reduced from 10 to 5 h by doubling the membrane area while maintaining the same flux. This would increase the purification raw material costs (Table 3), as the VRF membrane is one of the largest raw material costs. However, membrane reuse would counter this cost increase. The doubling of the VRF step membrane area would allow the entire process to fit within 24 h. In theory, a single purification train could therefore process a set of 12 bioreactors, for a plant capacity of 20 tons per year.

In a processing scenario where there is sufficient assurance of the segregation of individual batches of product due to the one-way product flow through the various rooms of the purification process suite, the total capacity of the purification process could be dictated by the longest individual unit operation. (This type of facility is in use in cGMP manufacturing plants such as insulin, where the large number of unit operations in the purification process train dictate the processing of several batches simultaneously.) The capacity of this single purification train is actually much higher than 10 tons per year if the restriction of processing only a single batch at a time can be relaxed! Table 5 summarizes several options to enable processing higher titer harvests that could capitalize on alternate management scenarios of the purification train. Since no single purification step has a duration even close to the 48 h harvest cycle, higher titer harvests could be processed if multiple 60-kg sub-batches were derived from a single bioreactor harvest, and the product stability in the harvested conditioned medium would allow multiple batches to be initiated from repeated transfers from the harvest tank to the ProA column within the 48 h harvest-to-harvest window. With a 10 h VRF step as the limiting unit operation (and assuming an additional 6 h for setup and CIP), three full processes could be completed in a 48 h period, offering a production capacity of 30 tons per year. Doubling the VRF membrane area while processing multiple batches in the suite would enable six sub-batches per 48 h period or 60 tons per year. Should a very high titer process be combined with a 20,000-L bioreactor, the reactor can be run at partial volume to match the output with the downstream process capacity. While these examples seem unrealistic, they illustrate the enormous capacity of the conventional unit operations used in the purification process to handle VLS processes without incurring unprecedented technical challenges. It is true that the close scheduling of so many sub-batches is probably not realistic and certainly a significant risk to a high batch success rate, but the thought exercise is useful to probe what is the true bottleneck of such a purification process.

6.2. Solution, Raw Material, and QC Testing Considerations. The two-column process described here has been

Table 5. Options for Handling Higher Titers or Shorter Bioreactor Cycles

Tons/ Year	Titer (g/L)	Harvest schedule (Days)	2nd suite	2 ProA cycles	2X VRF + 2 sub batches	> 1 batch per suite	AEX+VRF & concurrent UF/DF
10	5	12	Base case				
20	10	12	■				
				■			
					■		
	5	6	■				■
30	15	12				■	
40	20	12			Combined	Combined	
	10	6					■
60	30	12				Combined	Combined
					Combined	Combined	

developed with the goal of reducing the number of unit operations, as well as the solution number and volume necessary to support the purification process. The total number of solutions (excluding solutions for skid CIP/SIP that do not contact the resins or membranes) is only 13 for this platform process. Just 12 chemicals are needed as raw materials. One example of the steps taken to reduce the number of solutions needed is the use of a single equilibration solution for all chromatography and membrane filtration steps. The equilibration solution composition is determined by the pH and counterion concentration needed for the AEX step run in WPC mode. The ProA resin, VRF, and UF/DF modules can be equilibrated in the same solution without any impact on step performance. This minimizes the number of solutions and eases pressures on the solution preparation suite to meet precisely choreographed purification schedules. The reduction in the number of solutions needed will greatly simplify the solution preparation suite's activities for each batch.

Another metric used to characterize different processes is the number of liters of solution needed to purify 1 g of product. This purification process requires only 1.5 L per gram of mAb (note that this volume does not require the use of solution concentrates together with in-line dilution, which could further reduce the solution volume requirements). This is approximately one-third of the volume used by early clinical processes developed at Wyeth that still employed three column steps and had not undergone any efforts to intensify the process. For extant facilities that may have limited solution preparation capabilities, a decrease in solution volume of this magnitude can greatly increase the facility's capacity. In addition, a dilute process pool arising from a relatively low chromatographic column capacity, a modestly large elution pool, or a significant load dilution could introduce another large volume pool into the process flow, increasing downstream equipment size and thus impacting the required equilibration solution volumes and product pool tank volumes.

The reduction in raw materials for the two-column process will minimize the handling of large volumes of solid chemicals, easing pressures on the warehouse. Quality control intake testing is reduced in proportion to the number and volumes of raw materials. Raw materials are usually one of the most common sources of deviations requiring investigation in commercial manufacturing, and so a reduction in the number of raw materials needed could potentially reduce the risk of process upset and subsequent investigation. The reduction in the number of in-process pools decreases the total number of samples that are taken for in-process control testing for product concentration, endotoxin, bioburden, etc. This reduces the labor required for each batch, as well as providing a likely reduction in the number of investigations launched due to false positives arising from errors in sample handling.

7. Additional Processing Efficiencies

Additional processing efficiencies that may be afforded by alternate processing strategies for the two-column process are described here. A further reduction in the number of in-process pools is possible for the AEX and VRF steps. By plumbing the AEX column effluent line directly to the VRF filter, a single skid could be used to operate the AEX and VRF steps in tandem (see Figure 2B). Combined with a doubling of the VRF area to allow for a 5 h unit operation, the overall processing time of the purification train could be reduced to 18 h. If the UF/DF step could be started up once sufficient permeate volume from the AEX/VRF step had accumulated, the overlap of the 4 h UF/DF process time could allow the centrate to be completely processed to BDS in 12 h (see Figure 4B). These approaches would be another means of increasing the production capacity of the single purification suite, enabling 40 tons per year from a 6 day, 10 g/L cell culture process, or 60 tons per year from a 12-day, 30 g/L process (Table 5).

The number and size of in-process pool storage tanks could be a bottleneck for existing facilities. In the two-column process, this limit is extended by the reduction in the number of steps and the minimal dilution of the ProA pool through the subsequent steps. No intermediate UF/DF steps are needed to condition the load for the next unit operation, which is a further savings. The need to generate a well-mixed ProA pool suitable for the low pH virus inactivation requires this step to have a pool vessel or vessels capable of holding the entire pool. The use of 2,000-L disposable bags would require just two bags to be connected together to support the base case process.

Should the bioreactor titer double to 10 g/L, the option of handling 20 tons per year could be addressed by building out a second purification suite, as described above (this would also enable purification from a 6-day, 5 g/L titer cell culture process). Alternatively, some of the scenarios described in the section on scheduling could be used instead (see Table 5). Another option would be to cycle the 2-m ProA column twice, and then scale up all of the subsequent steps. The AEX column would only grow to 1.6 m and the membrane areas of the VRF and UF/DF steps would be doubled, which is still not an excessive scale for ultrafiltration operations. The in-process tankage requirements would grow proportionately to the increased batch size and probably require dedicated tanks. However, no technical challenges associated with the conventional purification unit operations would limit the production capacities. Even further increases in capacity could be envisaged, as in a two-suite operation with some of the scheduling options described in Table

5. There appear to be few drivers for alternate technologies in order to circumvent production capacity limitations.

8. Single 10-Ton Product vs Multiple Smaller Products

The need for 10 tons of cGMP mAb for therapeutic use is an unlikely situation. The clinical indication and global market would almost certainly require a large dose, chronic administration, a large patient population, and an unmet medical need with little competition. Examples of such indications could include cardiovascular protection, asthma/allergy, Alzheimer's disease, or stroke prevention. At this time, the maximum single dose for approved mAbs is 800 mg (the average dose for the 16 mAbs currently approved in the U.S. is closer to 150 mg). Assuming a 1,000 mg dose, the 10-ton production capacity would generate 10 million doses per year. Great strides have been made in increasing mAb affinity and half-life through protein engineering, and it is likely that a second-generation product would be developed to follow the lead mAb and alleviate the need for such an enormous production capacity due in part to a high product dose.

With an output of 10 million doses per year at 1 g per dose, monthly administration would allow treatment of 800,000 patients. This is very large patient population, accounting for 0.1% of the combined population of the United States, the EU and Japan. At a more reasonable 150 mg dose, this value rises to 0.6%, which is a very large fraction of the total population (roughly 60% of the total population of Alzheimer's patients in the U.S.). Few indications would command this number of patients.

At a sales price of $500 per gram, approximately the sales price of recombinant human insulin, the 10-ton process would be a $5 billion per year product. This would be among the top three biopharmaceutical products as of 2006. The COGs estimate detailed above would certainly support this sales price. Note that the sales price of Rituxan is approximately $4,000 per gram; 10 tons of Rituxan would retail for $40 billion, a very unlikely situation, as the combined health care systems of world could not support such an enormous outlay of the total health care budget. These analyses suggest that there will not be many situations that would require this massive annual production capacity of a mAb and that this scenario is unlikely to come to pass. If such an "Apollo program" is required, conventional technology can still deliver on the promise of high capacity and manageable production costs.

If the 10-ton capacity for a single product is unlikely, what is the future more likely to hold? The more probable scenario is that a facility having great production capacity by virtue of a combination of a large installed bioreactor volume and high titer cell culture processes will be campaigned between multiple products. As described above, there could be a benefit to expanding to a second purification suite to allow facile adjustment of output to changing market demands, by proportional scheduling bioreactors in a facility capable of running two products simultaneously. Alternatively, the single purification suite could be used to process one product at a time, and the entire facility used in a campaign mode to produce multiple products each year. If one assumes a changeover period of 1 month between products (a very conservative estimate), plus six weeks to build through the inoculum to the first harvest of a production bioreactor, the facility could probably produce three different products at capacities of 1 ton per year (see Figure 5). Each campaign would be satisfied by 16 batches, or 32 days of purification per product. Put another way, if the Rituxan and Enbrel processes employed cell culture processes with 5 g/L

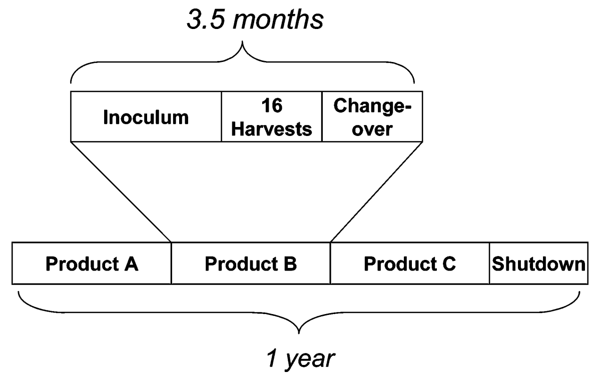

Figure 5. Annual production schedule for campaigning multiple products.

titers, current demands for both products could be met in 6 months, leaving the facility free to cover an additional blockbuster product in the same year and still have over a month for shutdown! A fourth process could fit into the calendar year if there were no shutdown, and the duration of the inoculum buildup could be trimmed a few weeks by using a rolling changeover of inoculum and production suites. If one of these process titers exceeds 5 g/L titer, one of the solutions described in Table 5 could allow that same campaign to produce more than 1 ton.

A short changeover period is key to this multi-product scenario and emphasizes the benefits of a single platform technology used for production of many products. The 10-ton mAb purification process described is based on a platform used at Wyeth for multiple cGMP mAb processes. The efficiencies in changeover between purification processes that use the same two-column platform are many-fold. In this case, the unit operations are the same, so the same equipment, core documentation, scheduling, and operations are used. Within a platform, there are no new raw materials needed for different processes. The warehouse will hardly know that a product changeover has occurred. Of the 13 process solutions needed for the mAb purification used as a design basis for the 10-ton process, a second product may need only four unique solutions, as at least nine are common between the two processes. These and other examples of the benefits of platform operation have become clear as many companies are now producing a series of mAbs in their clinical production facilities, which if they adhere to a platform approach, are enjoying the benefits of standardization.

For the case where one or more of the products is derived from a cell culture process with significantly lower titer, the purification train would require modification to reduce the scale of the batch size produced. This may entail smaller equipment, such as the final UF/DF skid. The purification production schedule, however, would not be greatly impacted, as only the ProA load time would be extended in order to load to the dynamic binding capacity of the column (which may be slightly reduced if the titer is markedly lower).

9. Implications for Nonconventional Unit Operations

Some alternative technologies being considered to replace conventional unit operations may have limitations that are revealed when a 10-ton process is considered. The value of considering this conceptual design is that it provides a clear comparison between current and potential technologies at the upper limit of production capability.

First, the scale-up of novel unit operations is uncertain. To be the first company to operate a new technology at 60-kg batch scale is, without question, a technical challenge. A $500 million dollar facility that is plagued by start-up delays will be a difficult situation, both for the corporation hoping to capitalize on an emerging market, and the patient population who await a therapy that is supplied from a single VLS facility.

Second, although small process yield losses are not consequential to the COGs (a 10% reduction in yield will give rise to only an 11% increase in COGs, which has already been shown to be rather small), they will cause an opportunity cost of lost production for a multi-product facility that will be much greater. In the case of an extended, inefficient facility changeover, each lost week of production is equivalent to 210 kg of product for the base case process.

Third, vendors who will provide raw materials for the novel processing step may also be scaling-up their production for the first time. The variability in raw material quality is an uncertain risk, and one that is difficult to manage. The disposal costs of process effluents containing high concentrations of salts, polymers, or excipient that would generate costly BOD or COD streams should be considered. These ancillary costs could be significant and may even restrict the flexibility of transferring the process to multiple production facilities where environmental controls could be different.

If the novel unit operation carries a modest royalty for the intellectual property covering this technology, this may become a significant cost and could factor strongly into the process economics. A 1% royalty on sales for a product selling for $500 per gram is $5 per gram, an amount greater than the direct materials COGs. For a mAb such as Rituxan, this 1% royalty would be $40 per gram, which can exceed the COGs of the entire drug substance and product processes.

Next, imagine a facility of this type operating in a campaign mode, but switching to a process that employs a non-conventional unit operation such as crystallization, EBA, or SMB. What additional delays will be incurred by the installation and validation of new equipment needed for the process? What specific training will be required for process operators to bring the operational success rate of the novel process up to the levels typical for chromatography or ultrafiltration? When the facility is switched back to a conventional process, a longer changeover period is likely as well. Short changeover periods maximize the production capabilities of the facility and capture the opportunity costs of down production time. The cost of a lost month (or two) of production arising from shifts in technology platforms can be very high indeed.

Finally, the novel process step may have “hidden” unit operations. Additional filtration, centrifugation, or other processing steps require equipment, product tankage, and solutions necessary to complete the unit operation. Each additional unit operation will have associated product losses, scaling challenges, and validation. These costs must be considered in comparison to the two-column platform process, which is used as the base-case of the 10-ton mAb purification train described here.

Regarding product purity levels, if two columns are still needed in addition to the novel unit operation in order to provide sufficient clearance of host cell proteins, nucleic acids, or product aggregates, then the simpler process using only two columns would appear to have significant advantages. If ProA chromatography is still used in the process, a polishing step that removes leached Protein A will likely still be needed. This analysis changes, of course, when the train containing the novel process step is compared to a three-column platform process, if the alternative technology is able to eliminate one of the two polishing steps.

In contrast to novel unit operations, the purification process described here has many advantages. The principles of the separation are well-established for centrifugation, pad filtration, ProA, and AEX chromatography, VRF and UF/DF steps. No process step requires equipment that exceeds current capacities; all of these conventional steps have been run at these scales in other facilities (the exception being the 2-m ProA column, but there are no known technical barriers to this operation).

It is noteworthy that the design basis for this process is a cGMP clinical mAb purification process that produced 10-kg batches. The BDS impurity levels were approximately 10-fold lower than typical Phase I/II targets (*29*), and the overall purification yield was 80%. For all unit operations, the scaling principles have been established. The definition of the process characterization and validation packages necessary for licensure is straightforward and unlikely to cause delays in the review of a license application. In comparison, the scale-up and process validation risks for novel technologies have to be higher.

Should the final production facility not become available in time for commercial launch, a bridging strategy employing a contract manufacturing organization (CMO) for BDS supply may be required. In other scenarios, a CMO may be needed to supplement the capacity of an existing facility on a temporary basis. Because the process described here does not involve any novel equipment or unit operations or unusual raw materials, the technology transfer and facility start-up will be simple in comparison to the case where unique unit operations are needed. (The production costs may be lower as well, reflecting the reduction in risk assumed by the CMO.)

When considering the clearance of adventitious virus by the purification process, the combination of low pH inactivation, AEX chromatography, and VRF should provide a sufficient package to ensure product safety. If novel unit operations are used and viral clearance is to be claimed, questions regarding the validity of the scale-down model may be challenging, especially for unit operations that are more likely to have scale dependencies.

10. Potential Improvements to the 10-Ton Process

There are several improvements to the process described here that might be considered for future versions of the process. Some of these could be implemented as post-licensure changes that may not require extensive comparability analysis or clinical trials.

The ProA step is designed based on currently available matrices. If the dynamic binding capacity could be increased from 50 to 80−100 g/L, significant benefits would be gained. The number of cycles per batch could be reduced, saving solution volume, labor, and process pool testing. The product concentration in the elution pool might be increased, which would help with the management of in-process pool volumes (provided the AEX and VRF steps could handle the elevated product concentrations).

There may be one further opportunity for step reduction. The use of expanded bed adsorption (EBA) could allow the harvest to be processed directly through an EBA-based ProA capture step, thus eliminating the centrifugation step and the associated pad filters. There might provide an increase in the overall process yields, as the centrifugation and pad filter losses would be eliminated. EBA appears to be a promising technology and one that is amenable to a platform technology. There are concerns, however, regarding the suitability of current EBA equipment and resins for VLS cGMP processes. In addition, the scheduling flexibility afforded by holding a cell-free clarified centrate may be lost if the whole cell-containing harvest must be held for 2 days.

Another area of improvement is the approach to reduce the extent of precipitation commonly observed in the centrate and neutralized ProA pool. Technologies such as flocculation of the CHO harvest prior to centrifugation have shown promise in lowering the pad filter area required for these two process pools (*33*). This will modestly reduce the purification costs and also decrease the amount of WFI needed in the facility, as flushing the pad filters requires a considerable volume of water.

Finally, the combination of AEX and VRF steps into a single operation, as described in Section 7, could have great value in reducing the footprint of the process, eliminating a process skid and in-process pool. It is likely that it will not take much effort to complete the development of this step and assemble the process validation package together with the necessary elements of the virus clearance validation.

11. Summary and Conclusions

This conceptual process design was based on a production platform used to purify a recombinant mAb secreted at high titers from mammalian cells. The conclusions regarding the suitability of conventional purification unit operations for VLS production should be restricted to this case.

The use of a highly efficient technology platform for mAb purification appears to offer enormous production capacity and modest direct raw material costs. The 10-ton production scale from a single purification train that was used for this conceptual design could actually be less than a quarter of the maximum capacity of the purification train if conservative improvements to the process steps were adopted, such as combining the AEX and VRF steps and running the plant 24/7.

The process intensification efforts that led to the step reduction and a two-column process were motivated by design principles that would enable VLS production. The philosophy of using conventional unit operations employing currently available resins and membranes will reduce scale-up, technology transfer, process robustness, and validation risks.

This conceptual design using an efficient platform purification train reflects the successful industrialization of the manufacturing processes for recombinant monoclonal antibodies. Unlike the chemical synthesis and cGMP production of small molecules, the same unit operations and process platform can be used for several different products. Many monoclonal antibody products will benefit from these efficiencies and make this class of biopharmaceutical products available to a much larger patient population, through both cost management and increased production capacity.

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References and Notes

(1) Jagschies, G.; Grönberg, A.; Björkman, T.; Lacki, K.; Johansson, H. J. Technical and economical evaluation of downstream processing options for monoclonal antibody (MAb) Production. *Suppl. BioPharm Int*. **2006**, Jun 2.

(2) Luan, Y-T.; Wang, W.; Nolan, R.; Drapeau, D. Defined Medium Development for High Yielding Mammalian Cell Culture Processes. IBC & Bioprocess International Conference; 8 November, 2006; San Francisco, CA.

(3) Low, D; O'Leary, R.; Pujar, N. S. Future of antibody purification. *J. Chromatogr. B* **2007**, *848*, 48−63.

(4) Curling, J.; Bryant, C. The plasma fractionation industry new opportunities to move forward? *BioProcess Int.* **2005**, *3* (3), 10−16.

(5) Smith, E. M. Chromatographic Albumin Production−Pilot to Production. *Downstream* **2000**, *31*, 6−8.

(6) Johnston, A.; Adcock, W. The Use of chromatograph to manufacture purer and safer plasma products. *Biotechnol. Gen. Eng. Rev.* **2000**, *17*, 37−70.

(7) Martin, T. IGIV: Contents, properties, and methods of industrial production−evolving closer to a more physiologic product. *Int. Immunopharmacol.* **2006**, *6*, 517− 522.

(8) Curling, J. Affinity chromatography−From textile dyes to synthetic ligands by design. *BioPharm* **2004**, *17* (7), 34−42.

(9) Ladisch, M. R.; Kohlmann, K. L. Recombinant human insulin. *Biotechnol. Prog.* **1992**, *8*, 469−478.

(10) Li, F.; Zhou, J. X.; Yang, X.; Tressel, T.; Lee, B. Current therapeutic antibody production and process optimization. *Bioprocess. J.* **2005**, *Sep/Oct*, 23−30.

(11) Shukla, A.; Hubbard, B.; Tressel, T.; Guhan, S.; Low, D. Downstream processing of monoclonal antibodies−Application of platform approaches. *J. Chromatogr. B* **2007**, *848*, 28−39.

(12) Kempken, R.; Preissmann, A.; Berthold, W. Assessment of a disc stack centrifuge for use in mammalian cell separation. *Biotechnol. Bioeng.* **1995**, *46*, 132−138.

(13) Bender, J.; Wolk, B. *Putting a Spin on CHO harvest: Centrifuge technology development*, National Meeting of the American Chemical Society, Boston, MA, August, 1998; American Chemical Society: Washington DC, 1998.

(14) Hahn, R.; Schlegel, R.; Jungbauer, A. Comparison of Protein A affinity sorbents. *J. Chromatogr.* **2003**, *790*, 35−51.

(15) Swinnen, K.; Krul, A.; Van Goidsenhoven, I.; Van Tichelt, N.; Roosen, A.; Van Houdt, K. Performance comparison of Protein A affinity resins for the purification of monoclonal antibodies. *J. Chromatogr. B* **2007**, *848*, 97−107.

(16) Brorson, K.; Krejci, S.; Lee, K.; Hamilton, E.; Stein, K.; Xu, Y. Bracketed generic inactivation of rodent retroviruses by low pH treatment for monoclonal antibodies and recombinant proteins. *Biotechnol Bioeng*. **2003**, *82*, 321−329.

(17) Kelley, B.; Tobler, S. A.; Brown, P.; Coffman, J.; Godavarti, R.; Iskra, T.; Switzer, M.; Vunnum, S. Principles of weak partitioning chromatography for anion exchange purification of monoclonal antibodies. *Biotechnol. Bioeng.* **2007**, submitted for publication.

(18) Fahrner, R. L.; Knudsen, H. L.; Basey, C. D.; Galan, W.; Feuerhelm, D.; Vanderlaan, M.; Blank, G. S. Industrial purification of pharmaceutical antibodies: Development, operation, and validation of chromatography processes. *Biotechnol. Genet. Eng. Rev.* **2001**, *18*, 301−327.

(19) Carter, J.; Lutz, H. An overview of viral filtration in biopharmaceutical manufacturing. *Eur. J. Parenter. Sci*. **2002**, *7* (3), 72−78.

(20) Phillips, MW; Bolton, G; Krishnan, M; Lewnard, JJ; Raghunath, B. Virus filtration process design and implementation. In *Process Scale Bioseparations for the Biopharmaceutical Industry*; Shukla, A. A., Etzel, M. R., Gadam, S., Eds.; Taylor & Francis: Boca Raton, 2007; pp 333−365.

(21) Bolton, G.; Cabatingan, M.; Rubino, M.; Lute, S.; Brorson, K.; Bailey, M. Normal-flow virus filtration: Detection and assessment of the endpoint in bioprocessing. *Biotechnol. Appl. Biochem.* **2005**, *42*, 133−142.

(22) Hongo-Hirasaki, T; Yamaguchi, K; Yanagida, K; Okuyama, K. Removal of small viruses (parvovirus) from IgG solution by virus removal filter Planova®20N. *J. Membr. Sci.* **2006**, *278*, 3−9.

(23) Williams, A.; Taylor, K.; Dambuleff, K.; Persson, O.; Kennedy, R. M. Maintenance of column performance at scale. *J. Chromatogr. A* **2002**, *44*, 69−75.

(24) *Scaled-up Method for Packing MabSelect in Large Diameter Chromaflow Columns*; Technical Brief 11-0007-52 B; Amersham BioSciences (now GE Healthcare): Uppsala, Sweden, 2004; 2004-09.

(25) Kelley, B. D. Establishing process robustness through designed experiments. In *Biopharmaceutical Process Validation*; Zabriske, D., Sofer, G., Eds.; Marcel Dekker: New York, 2000; pp 29−59.

(26) Brorson, K.; Brown, J.; Hamilton, E.; Stein, K. E. Identification of Protein A media performance attributes that can be monitored as

surrogates for retrovirus clearance during extended re-use. *J. Chromatogr. A* **2003**, *989*, 155−163.

(27) Farid, S. S. Process economics of industrial monoclonal antibody manufacture. *J. Chromatogr. B* **2007**, *848* (8−18).

(28) Chadd, H. E.; Chamow, S. M. Therapeutic antibody expression technology. *Curr. Opin. Biotechnol.* **2001**, *12*, 188−194.

(29) Clearer Paradigm Sought For Biopharmaceutical Specifications. *Gold Sheet* **2004**, *38* (2).

(30) Farrugia, A.; Robert, P. Plasma protein therapies: current and future perspectives. *Best Pract. Res. Clin. Haematol.* **2006**, *19*, 243−58.

(31) Blank, G. S. Design and Scale-up of a Monoclonal Antibody Recovery Process. Recovery of Biological Products X; 7 June, 2001; Cancun, Mexico.

(32) Madani, H. Enbrel Product Lifecycle from Phase I to Market Lauch and Beyond. Recovery of Biological Products XI; 18 September, 2003; Banff, Canada.

(33) Shpritzer, R.; Vicik, S.; Orlando, S.; Acharya, H.; Coffman, J. L. Calcium phosphate flocculation of antibody-producing mammalian cells at pilot scale. *Abstracts of Papers*, 232nd National Meeting of the American Chemical Society, San Francisco, CA, Sept 10−14, 2006; American Chemical Society: Washington, DC, 2006; BIOT-080.

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